

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		Ticks in boxes 1, 5 and 6 (3) Television 1 uses less energy per second than television 2 ✓ Television 3 uses 40 units more per year than television 4 ✓ Televisions with the same energy rating, e.g A++, don't always have the same power ✓ –1 mark for each additional box ticked		3		3		
	(b)			N.B. Only televisions 1 and 2 to be used. 1st mark – correct substitution of one ratio 2nd mark – correct calculation of one ratio 3rd mark – correct calculation of 2nd ratio 3 marks to be awarded only if correct conclusion present Screen size to screen size compared with power to power $\frac{139}{69} = 2.01$ $\frac{78}{32} = 2.44$ OR $\frac{69}{139} = 0.50$ $\frac{32}{78} = 0.41$ Conclusion – [Ratios not the same] so not true Alternative Ratio of screen size to power compared $\frac{69}{32} = 2.16$ $\frac{139}{78} = 1.78$ OR $\frac{32}{69} = 0.46$ $\frac{78}{139} = 0.56$ Conclusion – [Ratios not the same] so not true Alternative Ratio of screen size to kWh per year compared $\frac{69}{47} = 1.47$ $\frac{139}{108} = 1.29$ OR			3	3	3	